Appendix: Ramanujan Graph Rewiring with Non-Negative Resistance Curvature

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1 Appendix: Proofs of Sections 4 and 5

1.1 Proof of Theorem 1: Commute Time Bound via Resistance Curvature

Theorem 1 (Commute time bound via resistance curvature). *Let $G = (V, E)$ be a finite, connected, undirected, and unweighted graph. For any $u, v \in V$, let*

$$u = x_0 \sim x_1 \sim \cdots \sim x_k = v$$

be a shortest path, so that $k = d_G(u, v)$. Define

$$p_\star := \min_{0 \le i \le k} p(x_i), \qquad \textit{where} \qquad p(x) = 1 - \frac{1}{2} \sum_{y \in N(x)} \mathbf{Reff}(x, y).$$

Then

$$\mathbf{CT}(u, v) \le 4\,|E|\,k\,\bigl(1 - p_\star\bigr).$$

Proof. For the simple random walk on G, the commute time satisfies

$$\mathbf{CT}(u, v) = 2|E|\,\mathbf{Reff}(u, v), \tag{1}$$

see [2]. Moreover, $\mathbf{Reff}$ defines a metric on V, and therefore satisfies the triangle inequality. Applying this inequality along the shortest path

$$u = x_0 \sim x_1 \sim \cdots \sim x_k = v$$

gives

$$\mathbf{Reff}(u, v) \le \sum_{i=0}^{k-1} \mathbf{Reff}(x_i, x_{i+1}). \tag{2}$$

We now bound each term $\mathbf{Reff}(x_i, x_{i+1})$ through the resistance curvature. By definition,

$$1 - p(x) = \frac{1}{2} \sum_{y \in N(x)} \mathbf{Reff}(x, y).$$

Since all terms in the sum are nonnegative, for every neighbor $y \in N(x)$ we have

$$\mathbf{Reff}(x, y) \leq \sum_{z \in N(x)} \mathbf{Reff}(x, z) = 2\big(1 - p(x)\big). \tag{3}$$

In particular, for each edge (x_i, x_{i+1}) on the path,

$$\mathbf{Reff}(x_i, x_{i+1}) \leq 2\big(1 - p(x_i)\big).$$

By definition of

$$p_\star = \min_{0 \leq i \leq k} p(x_i),$$

we have $p(x_i) \geq p_\star$ for all i, and hence

$$1 - p(x_i) \leq 1 - p_\star.$$

Therefore,

$$\mathbf{Reff}(x_i, x_{i+1}) \leq 2(1 - p_\star), \qquad \text{for all } i = 0, \dots, k-1.$$

Combining this estimate with (2), we obtain

$$\mathbf{Reff}(u, v) \leq \sum_{i=0}^{k-1} \mathbf{Reff}(x_i, x_{i+1}) \leq \sum_{i=0}^{k-1} 2(1 - p_\star) = 2k(1 - p_\star).$$

Substituting this into (1) yields

$$\mathbf{CT}(u, v) = 2|E|\,\mathbf{Reff}(u, v) \leq 2|E| \cdot 2k(1 - p_\star) = 4|E|k(1 - p_\star),$$

which concludes the proof.

1.2 Proof of Theorem 2: Ramanujan Graphs Have Non-Negative Resistance Curvature

Theorem 2. *Let G be a Ramanujan graph on N vertices of degree d, and let $C > 0$ be the absolute constant from [6]. If*

$$d \geq 4C^2 (\log N)^{2/3},$$

then G has non-negative resistance curvature.

Proof. By the Ramanujan property,

$$\mu(G) \leq 2\sqrt{d-1} < 2\sqrt{d},$$

and therefore

$$\frac{d}{\mu(G)} > \frac{\sqrt{d}}{2}.$$

By [6], any (N, d, μ)-graph satisfying

$$\frac{d}{\mu(G)} \geq C(\log N)^{1/3}$$

is Hamiltonian. Moreover, Hamiltonian graphs have non-negative resistance curvature by [5, 4]. Hence, it is sufficient to require

$$\frac{\sqrt{d}}{2} \geq C(\log N)^{1/3},$$

which is equivalent to

$$d \geq 4C^2(\log N)^{2/3}.$$

This proves the claim.

1.3 Proof of Corollary 1: Low Diameter

Corollary 1. *Let G be a finite, connected Ramanujan graph on N vertices of degree d with non-negative resistance curvature. Then*

$$\operatorname{diam}(G) \leq 1 + \frac{3\,\log N}{\log(\log N)}.$$

Proof. Let G be a connected d-regular non-bipartite graph on N vertices, and let μ denote its second-largest adjacency eigenvalue. By Chung's adjacency-power bound [3],

$$\operatorname{diam}(G) \leq 1 + \frac{\log(N-1)}{\log(d/\mu)}.$$

If G is Ramanujan, then $\mu \leq 2\sqrt{d-1}$, and thus

$$\operatorname{diam}(G) \leq 1 + \frac{\log(N-1)}{\log\left(\dfrac{d}{2\sqrt{d-1}}\right)}.$$

By Remark 1 of Theorem 2 , we have $d \geq 4(\log N)^{2/3}$. Hence,

$$\frac{d}{2\sqrt{d-1}} \geq \frac{4(\log N)^{2/3}}{2\sqrt{4(\log N)^{2/3}-1}} \geq \frac{4(\log N)^{2/3}}{2 \cdot 2(\log N)^{1/3}} = (\log N)^{1/3}.$$

Taking logarithms gives

$$\log\left(\frac{d}{2\sqrt{d-1}}\right) \geq \frac{1}{3}\log(\log N).$$

Therefore,

$$\operatorname{diam}(G) \leq 1 + \frac{\log(N-1)}{\frac{1}{3}\log(\log N)} = 1 + \frac{3\,\log(N-1)}{\log(\log N)}.$$

Since $\log(N-1) \leq \log N$, we conclude

$$\operatorname{diam}(G) \leq 1 + \frac{3\,\log N}{\log(\log N)}.$$

1.4 Proof of Corollary 2: Maximum Effective Resistance

Corollary 2. *Let G be a Ramanujan RN graph of degree d on N vertices. Then the maximum effective resistance $R_{\max}$ satisfies*

$$R_{\max} \leq \frac{2}{d - 2\sqrt{d-1}}.$$

Proof. Combining the general bounds from [2, 1], we have

$$\frac{1}{N\lambda_2} \leq R_{\max} \leq \frac{2}{\lambda_2},$$

where λ_2 denotes the spectral gap. Since G is Ramanujan,

$$\lambda_2 = d - \mu_2 \geq d - 2\sqrt{d-1}.$$

Therefore,

$$R_{\max} \leq \frac{2}{\lambda_2} \leq \frac{2}{d - 2\sqrt{d-1}}.$$

This concludes the proof.

2 Description of graph classification datasets

In this section we report the main structural statistics of the graph classification datasets used in our experiments, as they help contextualize the behavior of the rewiring procedure across benchmarks with different graph sizes and densities.

Dataset	**#Graphs**	**#Nodes**	**#Edges**	$\overline{\mathbf{d}}$	Std(**d**)
MUTAG	188	17.9 (±5)	19.8 (±6)	2.2	0.7
PROTEINS	1113	39.1 (±46)	72.8 (±85)	3.7	0.9
ENZYMES	600	32.6 (±15)	62.1 (±25)	3.9	1.0
REDDIT	2000	429.6 (±554)	497.8 (±623)	2.3	8.9
COLLAB	5000	74.5 (±62)	2457.2 (±6439)	37.4	11.8
IMDB	1000	19.8 (±10)	96.5 (±105)	8.9	2.8

Table 1: Graph-level datasets: average structural statistics across graphs (mean ± standard deviation). $\overline{d}$, Std(d), denote mean, standard deviation.

- **Molecular datasets.** MUTAG contains small molecular graphs: nodes represent atoms, edges represent chemical bonds, and node labels encode atom types. PROTEINS and ENZYMES represent protein structures: nodes are secondary-structure elements with physicochemical annotations, and edges connect elements that are adjacent in the amino-acid sequence or spatially close in 3D. In PROTEINS, graph labels are binary. In ENZYMES, graphs are classified into the six top-level EC classes.

– **Social datasets.** IMDB-BINARY is constructed from actor collaboration graphs: nodes represent actors/actresses, and an edge indicates that two actors co-appeared in a movie. Each graph is assigned one of two movie-genre labels. COLLAB is built from researcher ego-networks, where nodes are researchers and edges represent collaboration links; graph labels correspond to research fields. REDDIT-BINARY is derived from discussion-thread interaction graphs, in which nodes are users and edges denote reply interactions; each graph label indicates the discussion community type.

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